

PROTOTYPE BLUEPRINT

Privacy-preserving visual perception for browser agents

A compact engineering definition of the system to be built, the mechanism that connects its parts, and the evidence required to call the prototype complete.

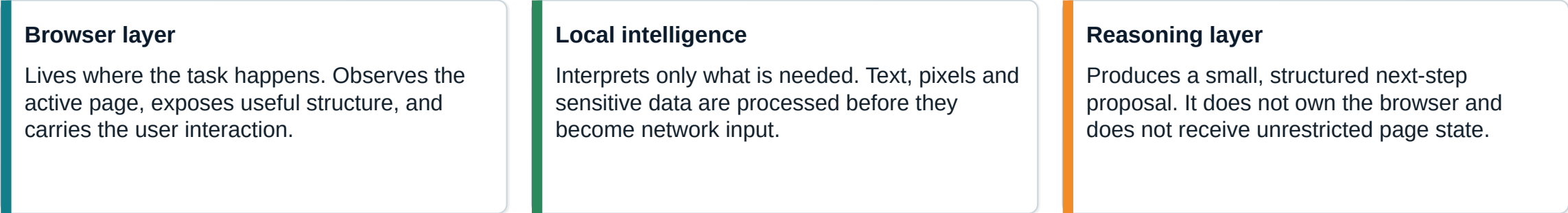


PROTOTYPE NORTH STAR

The browser keeps sensitive context and execution authority local; remote reasoning receives only the information it needs.

What the prototype is

One browser-resident system; one controlled reasoning path; one local execution authority.



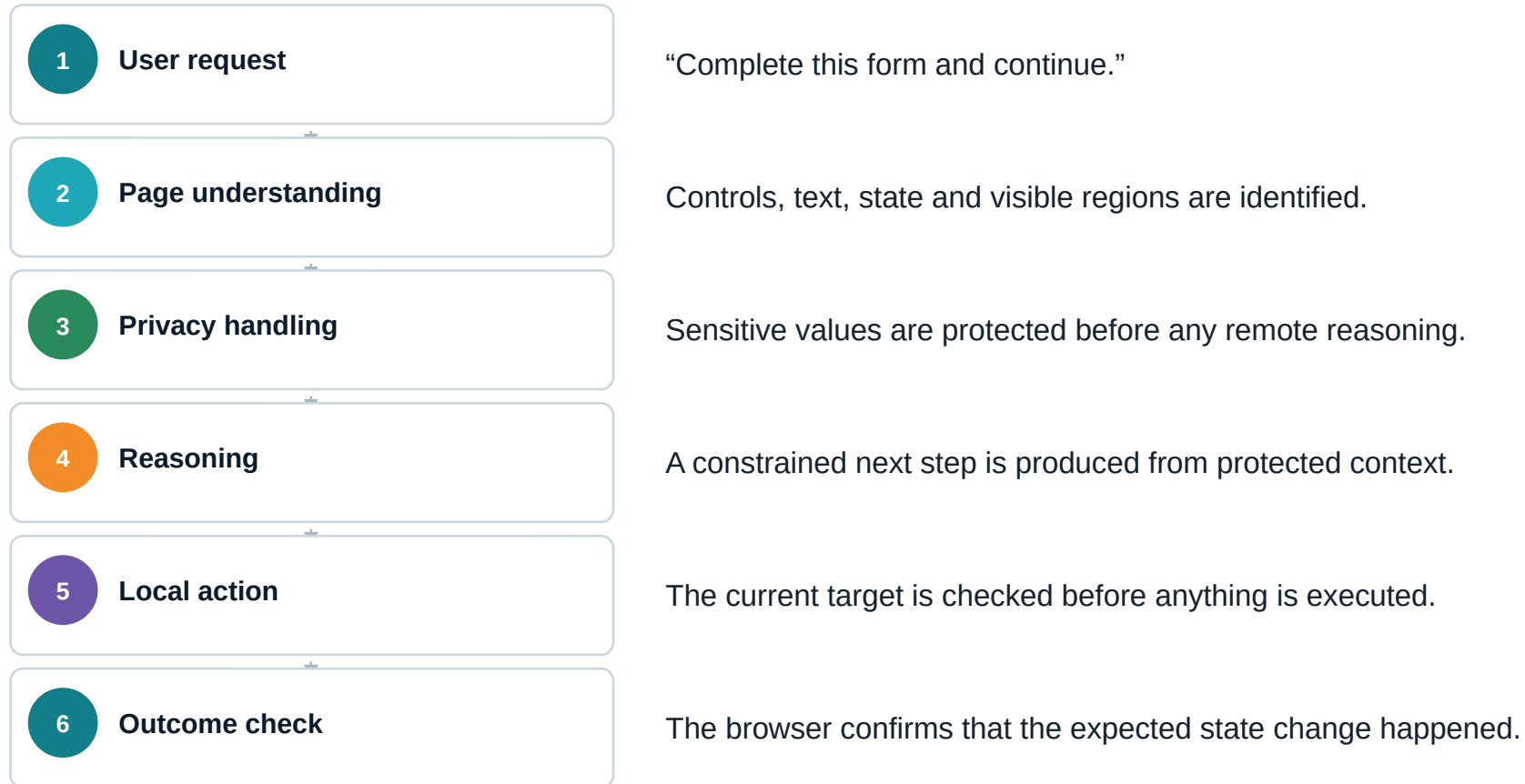
SYSTEM CONTRACT



Prototype boundary
The prototype is a complete vertical slice, not a collection of separate AI demos: observe -> protect -> reason -> control -> verify.

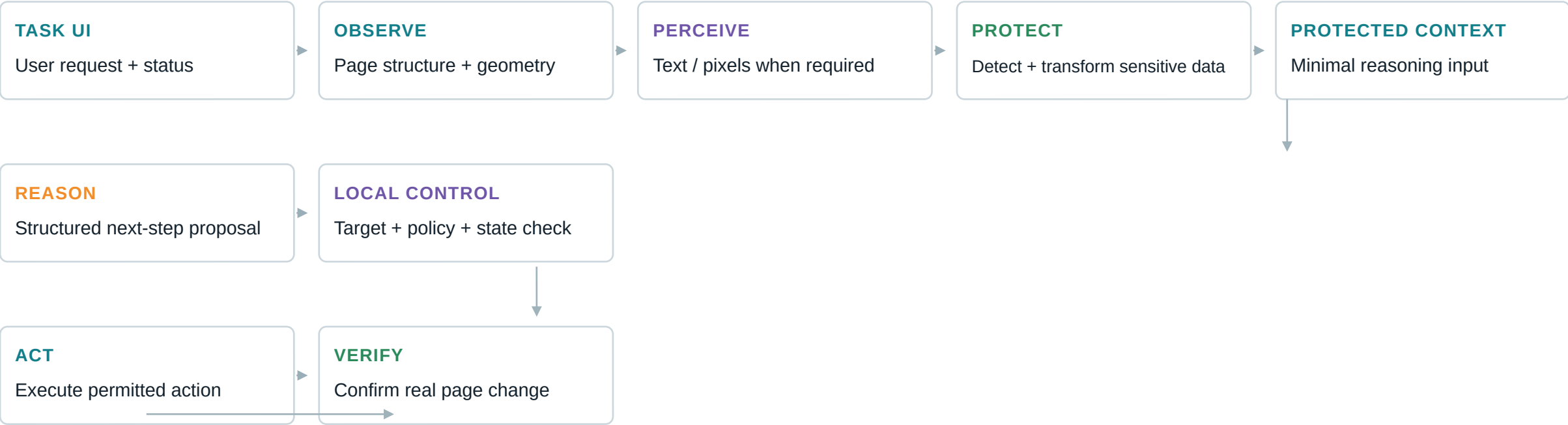
One task, end to end

The mechanism is easiest to understand when the system is shown doing one real job.



The complete prototype path

Every stage has a single job and a controlled handoff.



CORE PROPERTY Reasoning may influence the next action. It never bypasses local control.

Where each responsibility lives

A clean separation keeps the prototype understandable, testable and safe.

IN THE BROWSER TAB

1. Observe visible/interactable page state
2. Maintain live target references
3. Perform approved browser actions
4. Confirm post-action state

LOCAL PROCESSING

1. Interpret pixels only when needed
2. Detect and transform sensitive data
3. Construct protected task context
4. Apply local policy

REMOTE REASONING

1. Receive protected task context
2. Choose a next step
3. Return a constrained proposal
4. No direct browser control

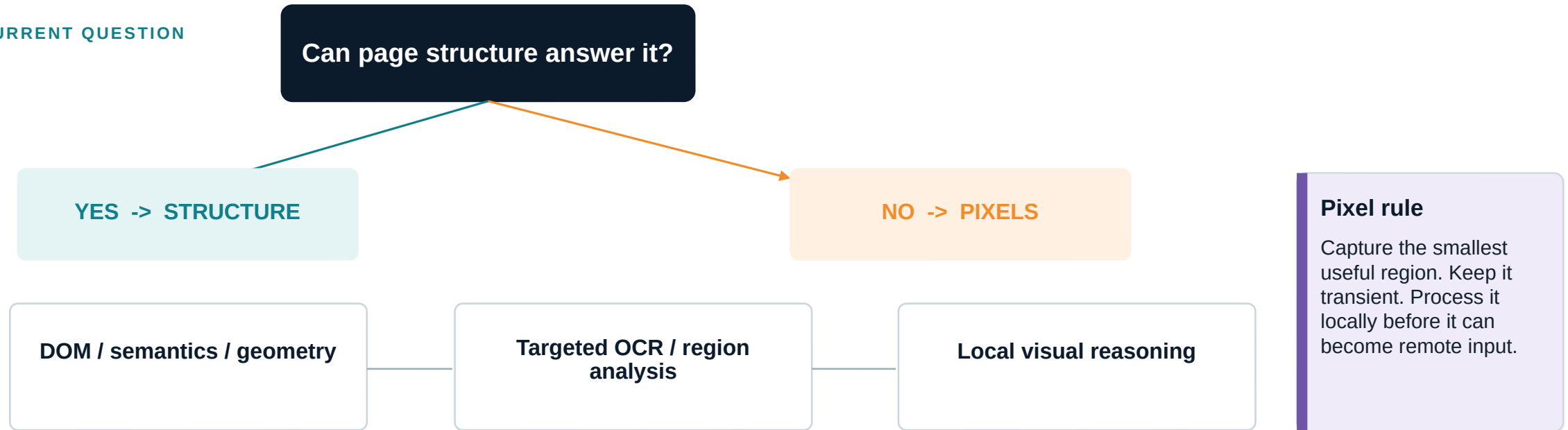
DATA RULE

Sensitive values and direct execution authority remain on the local side of the boundary.

The visual understanding strategy

Use the simplest reliable signal first; only add heavier perception when the page requires it.

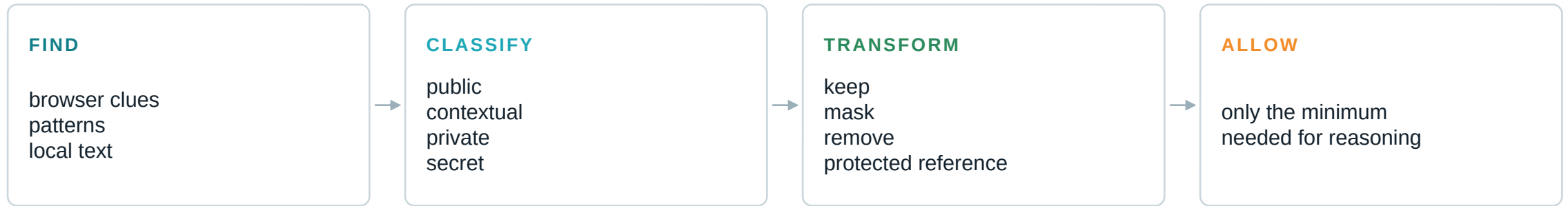
CURRENT QUESTION



Result: one perception system with multiple sensors, not one heavy model running on everything.

Protect useful context without exposing the original

The system separates “knowing that information exists” from “sending the information itself.”



ALWAYS LOCAL

Passwords • one-time codes • access/session tokens • equivalent secrets

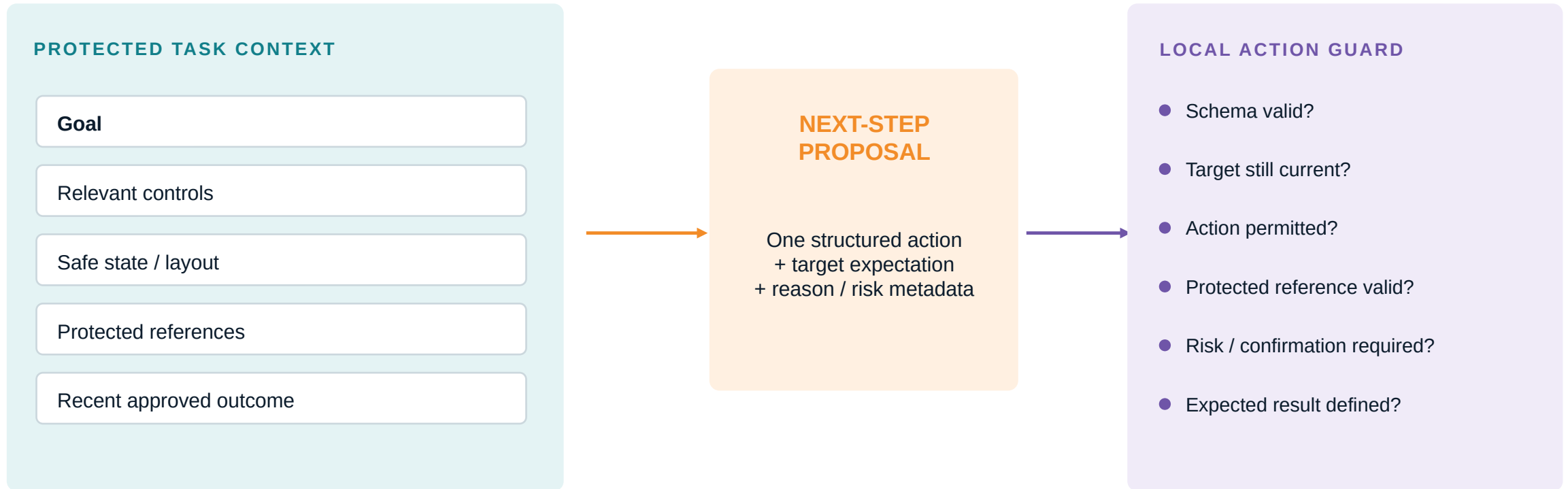
PROTECTED REFERENCE

A private value may be represented by a safe placeholder when its role is necessary for reasoning; the original remains local.

PRIVACY CHECKPOINT / The network boundary accepts protected context, not arbitrary page state.

Reasoning proposes; the browser decides

The remote component solves the “what next?” problem without owning the “may this action happen?” decision.



Only after the local checks pass does execution become possible.

Make actions reliable in a changing page

A planned target can become wrong in milliseconds. The prototype must validate the live page immediately before acting.



STALE / MISMATCH

Do not execute. Re-observe and rebuild the proposal from the current page.

VALID

Execute one action, then verify the expected state delta.

Verification signal

URL change • DOM/state change • expected control appears/disappears • success message • semantic state change

The system defines success by the page state, not by the model saying “done.”

The control surface

The interface exposes what matters: task state, protection state, reasoning state and action outcome.

Protected browser task

Application form

Email

[protected value]

Password

.....

Continue

N-EYE

TASK
Complete form

PROTECTION
2 values protected

REASONING
Next step ready

ACTION
Continue

RESULT
Verified

01 Focused

The interface exposes the path, not the implementation internals.

02 Legible

Every state has one obvious meaning and one next action.

03 Honest

Prototype state is visible; unsupported claims are not implied.

The prototype is defined by proof, not screenshots

Each major claim has a direct observable test.

CLAIM	WHAT MUST BE TRUE	HOW TO PROVE IT
Privacy	Protected values stay out of outbound requests	Seed canaries; inspect request body, URL, headers and logs
Visual perception	At least one target exists only in pixels	Use a visual-only test page; compare DOM-only vs perception path
Action safety	Unsafe/stale proposals are blocked	Change page state between planning and execution
Verification	Success is tied to actual page change	Define before/after state and expected delta
Performance	The path is efficient enough for the target machine	Measure cold/warm latency, payload size, memory proxy

RULE / If the test cannot distinguish success from failure, the requirement is not yet measurable.

The prototype ships with its own hostile test world

Controlled pages expose the exact failure modes the open web can hide.

01

Basic form

email / phone / password / action

02

Image text

PII only visible in pixels

03

Canvas control

actionable visual target

04

Dynamic page

target replaced after planning

05

Prompt injection

page text tries to override policy

06

Hidden data

sensitive content outside the visible task

07

Document view

text + region + icon-only controls

08

Failure path

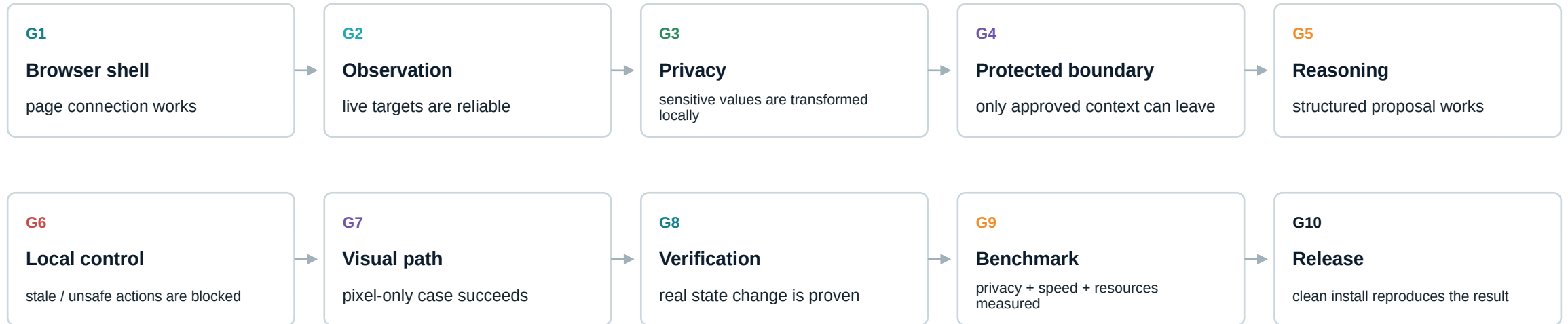
network / GPU / OCR / verification fault

HOLD-OUT RULE

Keep a subset of pages hidden from tuning. The prototype should succeed by mechanism, not by memorizing fixtures.

The prototype is assembled in gates

Each gate closes one risk before the next layer is allowed in.



DESIGN RULE

Do not add a sophisticated model to compensate for a broken boundary, target model or test system.

What is inside the first complete build

Enough to demonstrate the core mechanism; not padded with unrelated infrastructure.

Browser extension

Task surface, page observation, target handling

Local perception

DOM/semantics, targeted OCR, selected visual processing

Privacy layer

Detection, transformation, protected references, local policy

Reasoning service

Thin structured planner over protected context

Local control

Re-check, authorize, execute, verify

Test laboratory

Controlled pages, hold-outs, canaries, benchmark harness

OUT OF SCOPE FOR THE FIRST BUILD / broad multi-browser parity • large foundation-model training • elaborate microservices • full enterprise management

Prototype v0.1 is complete when...

- A real browser task can complete from start to finish.
- Sensitive values are transformed before remote reasoning.
- The visual path handles at least one pixel-only case.
- Remote reasoning returns a constrained next-step proposal.
- The browser rejects stale or unsafe actions locally.
- Execution is confirmed by an observed state change.
- Privacy, latency and resource measurements are reproducible.
- A clean environment can rebuild and reproduce the prototype.

CORE PRINCIPLE / Simple on the outside. Deep where it matters. Measurable before it is called finished.